

Evaluating "Use Instead of Curtail": An Interrupted Time Series Analysis of Germany's §13k EnWG Trial

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Introduction

Germany has a growing electricity problem that sounds strange at first: it sometimes has too much clean power. Wind and solar output in the windy north have grown faster than the transmission lines needed to carry that power south, to where most of it is used. When the grid can't handle the flow, grid operators have to switch wind and solar plants off temporarily a practice called curtailment and pay the plant owners for the electricity they weren't allowed to produce. This has become an increasingly expensive problem: more renewable capacity each year means more moments where supply outruns what the grid can carry.

In **October 2024**, Germany began testing a new tool to address this. Instead of simply switching renewable plants off, grid operators can now redirect the surplus power to nearby consumers who have signed up in advance such as battery storage systems or industrial heating units paying them to absorb the extra electricity instead of wasting it. This mechanism, known as **Section 13k** ("use instead of curtail"), is currently a two-year trial running in specific regions with the worst grid congestion, mainly in northern Germany.

This report asks a simple question: has the trial actually reduced curtailment? Answering it turns out to be harder than it sounds. Curtailment has continued to rise since the trial began, but so have several other things that have nothing to do with the trial Germany kept adding wind and solar capacity, and the country also shut down its **last nuclear power plants** in April 2023, which changed how electricity moves across the grid. Any honest answer has to separate the trial's effect from these other moving parts, rather than simply comparing "before" and "after" and assuming the difference is caused by the policy. This report uses publicly available grid data from Germany's transmission operators, covering the period from late 2021 through August 2026, to do exactly that.

§13k EnWG: "Use Instead of Curtail"

Germany's "use instead of curtail" mechanism was introduced through a November 2023 reform of the **Energy Industry Act (EnWG)** and formally took effect on 29 December 2023. It gives transmission system operators (TSOs) a second option when the grid becomes congested typically when wind output in the north exceeds what transmission lines can carry south instead of simply switching a renewable generator off and compensating it for the lost output. Under the provision, TSOs must publish an hourly forecast, by the morning of the previous day, of how much electricity is expected to require curtailment in designated "relief regions" areas identified as persistently congestion-prone. Eligible consumers registered in these regions, such as electrolyzers, battery storage, or power-to-heat systems, can then increase their consumption to absorb part of that forecast surplus, turning what would have been wasted electricity into productive use, and are paid for the additional load they take on. Following a June 2024 specification from the Federal Network Agency (BNetzA) on which technologies qualify, the TSOs' **two-year trial phase** began on **1 October 2024** under a simplified, flat-rate allocation procedure. That trial is scheduled to conclude on **30 September 2026**, after which the mechanism shifts to a revised framework based on daily competitive tenders meaning the period analyzed in this report covers essentially the entire lifespan of the **flat-rate trial**, right up to its scheduled handover to a different market design.

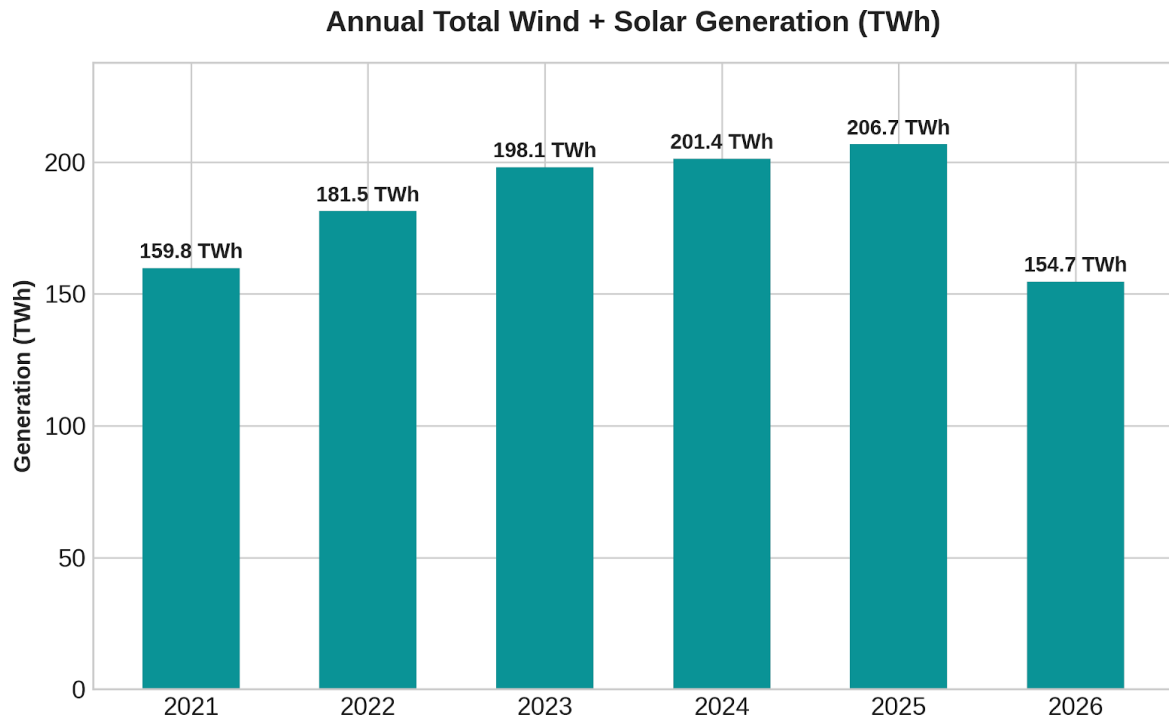


Figure 1 Annual Total Wind + Solar Generation, 2021–2026 (TWh)

Combined wind and solar output grew steadily from 159.8 TWh in 2021 to a peak of 206.7 TWh in 2025, reflecting continued capacity additions.

Note: the 2026 bar (154.7 TWh) covers only January–August and is not a full-year figure — shown alongside full years, it falsely appears as a sharp decline.

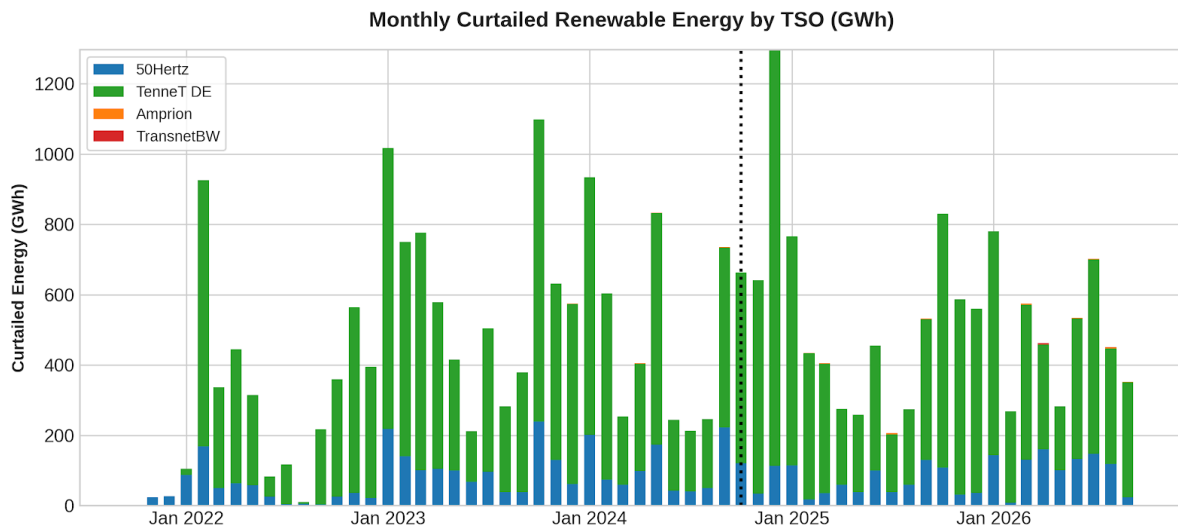


Figure 2 Monthly Curtailed Renewable Energy by Transmission System Operator (GWh)

Curtailment is heavily concentrated within TenneT DE's control area, which covers Germany's wind-dense northern regions, with 50Hertz contributing a smaller secondary share and Amprion/TransnetBW (covering western and southwestern Germany) contributing negligible volumes. This distribution is consistent with curtailment being driven primarily by north-south transmission congestion rather than a nationwide phenomenon.

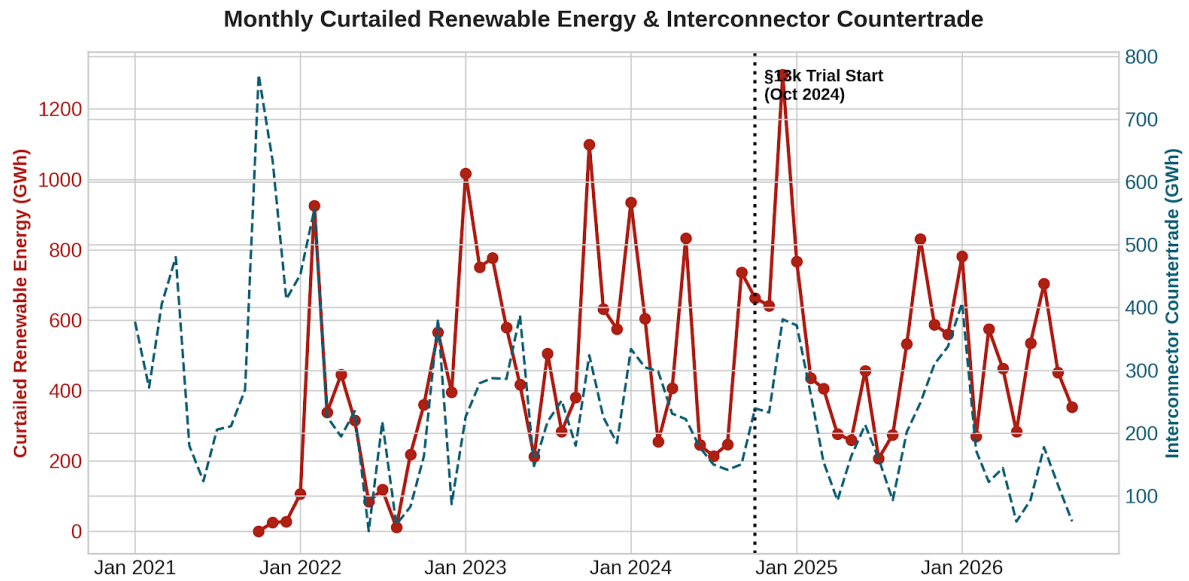


Figure 3 Monthly Curtailed Renewable Energy & Interconnector Countertrade (2021–2026)

This chart tracks Germany's monthly renewable curtailment (red, left axis) against cross-border interconnector countertrade volume (dashed teal, right axis), with the §13k trial's October 2024 start marked. Curtailment has trended upward with heavy month-to-month volatility since 2021, without a visually clear break at the trial start date.

Note: the earliest data point (October 2021) reflects an incomplete first month of source reporting.

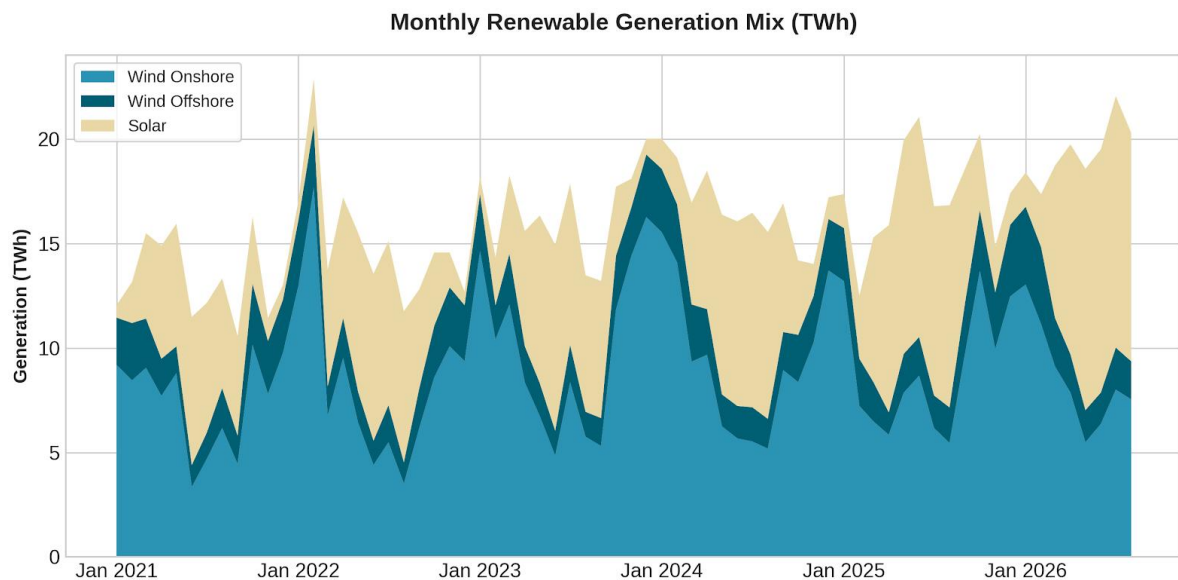


Figure 4 Monthly Renewable Generation Mix by Technology (TWh)

Wind onshore generation forms the base of Germany's renewable output, supplemented by a smaller offshore contribution and by solar, which follows a pronounced seasonal cycle — peaking each summer and falling to near zero in winter months. The combined stack shows a gradual increase in total renewable output over the period, consistent with continued capacity expansion.

Read **naively**, these figures above suggest curtailment rose after the trial began. That comparison is **unreliable**, however, because it ignores growing renewable output, strong seasonality, Germany's April 2023 nuclear phase-out, and the autocorrelation typical of a time series. To separate a §13k effect from these confounding factors, we estimate **an interrupted time series model** at weekly resolution, controlling for wind and solar generation, seasonality, and the nuclear phase-out, and we test the design with **placebo treatment** dates.

Table 1: Effects of Regulation on Wind and Solar Curtailment

Model: <i>Variables</i>	Dependent Variable: log_curtailed		
	(1) Full sample	(2) Trimmed sample	(3) + Nuclear control
	(1)	(2)	(3)
Time trend (weeks)	0.0128*** (0.0031)	0.0089*** (0.0023)	0.0375*** (0.0063)
Post §13k EnWG (level shift)	-0.5612* (0.2986)	-0.4510 (0.2808)	0.0101 (0.2916)
Weeks since §13k × slope change	-0.0143*** (0.0051)	-0.0087** (0.0040)	-0.0022 (0.0059)
log(Wind + Solar generation)	2.915*** (0.4303)	2.829*** (0.4048)	2.978*** (0.3656)
Post nuclear phase-out (level)			-0.9137*** (0.3455)
Weeks since nuclear × slope			-0.0362*** (0.0079)
<i>Fixed-effects</i>	Yes	Yes	Yes
month_of_year	251	248	248
Observations R^2	0.44644	0.47881	0.54034
Adjusted R^2	0.41110	0.44512	0.50637

Heteroskedasticity-robust (HC1) standard errors in parentheses.

All models include month-of-year fixed effects.

Column (2) drops weeks before 15 Nov 2021 (data-quality outliers).

Column (3) additionally controls for the final nuclear phase-out (15 Apr 2023).

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 1 reports the results of this specification across three progressively controlled models. We estimate an interrupted time series (ITS) model at weekly resolution ($n = 248$ weeks, November 2021 to August 2026), regressing log curtailed renewable energy on a linear time trend, a post-treatment level indicator, a post-treatment slope term, log wind-and-solar generation, and month-of-year fixed effects. Before controlling for Germany's April 2023 nuclear phase-out, the model shows a **borderline-significant post-treatment level drop** ($\beta = -0.451$, $p = 0.110$ under heteroskedasticity-robust errors; $p = 0.154$ – 0.204 under Newey-West HAC corrections at 4- and 8-week lags) and a **modestly significant slowdown in the post-treatment growth rate** ($\beta = -0.0087$, $p = 0.032$ to 0.073 across the same specifications). A placebo test using four counterfactual treatment dates found that three of the **four produced comparably or more significant "effects" than the real October 2024 date**, indicating the design was not cleanly isolating a treatment-specific break. Once we add controls for the **nuclear phase-out** (level and slope terms at April 2023), both §13k coefficients collapse toward zero and **lose all significance** (post-13k: $\beta = 0.010$, $p = 0.972$; slope change: $\beta = -0.002$, $p = 0.714$), while the **nuclear phase-out terms are themselves highly significant** ($p = 0.009$ and $p < 0.001$) and model fit improves substantially (adjusted R^2 rises from 0.445 to 0.506). Re-running the placebo test with the nuclear control held constant eliminates the significant level-shift at three of the four placebo dates, but a significant slope-change effect persists at three of the four, including the fully unresolved April 2024 date, indicating at least one additional unmodeled structural break. Across every specification, log wind-and-solar generation is a strong, consistently significant predictor of curtailment ($p < 0.001$ throughout), confirming **that renewable output growth is a substantial driver of the observed trend**. Taken together, these results provide **no robust evidence** that the §13k trial phase has measurably reduced renewable curtailment once the confounding effects of renewable capacity growth and the nuclear phase-out are properly accounted for.

Limitations

1. Scope of Curtailment Data

The dataset is drawn exclusively from netztransparenz.de's TSO-level redispatch records. While these data capture curtailment ordered directly by transmission system operators, they exclude distribution-grid-level curtailment (*Einspeisemanagement*). Consequently, the total curtailment volumes in this study are lower than officially reported national totals, and the analysis characterizes TSO-driven curtailment specifically rather than aggregate renewable curtailment across Germany.

2. Structural Correlation and Time-Series Constraints

The identification strategy relies on a single national time series with a single treatment date. Across every specification tested, this produces a strong structural correlation (≈ 0.85) between the post-treatment indicator and the underlying time trend. This represents an inherent methodological constraint in isolating the single-date policy effect from Germany's broader macro trajectory, regardless of additional control variables.

3. Unexplained Structural Breaks (Placebo Test Results)

Placebo testing across four counterfactual dates revealed that two remained statistically significant even after controlling for the April 2023 nuclear phase-out. This points to at least one additional, unidentified structural break in the time series. An investigation into potential policy or grid events around April 2024 yielded inconclusive results.

4. Binary Treatment Indicator vs. Gradual Ramp-Up

The treatment variable is modeled as a binary indicator marking the formal start of the trial. This specification cannot capture the gradual, continuous ramp-up in registered flexible-load participation inherent to the program's design. As a result, an effect that scales with participation over time may be understated or obscured by a level-and-slope specification tied to a single start date.

5. Aggregation of Wind and Solar Generation

Wind and solar generation are combined into a single control variable, which may mask technology-specific drivers. Wind curtailment is primarily driven by transmission grid congestion, whereas solar curtailment increasingly stems from midday oversupply and negative spot market pricing.

6. Granularity of Seasonal Controls

Seasonality is addressed using month-of-year fixed effects rather than finer week-of-year controls. While this was a deliberate choice to prevent overfitting given the limited sample size (≈ 250 observations), it leaves some short-term volatility and residual outlier weeks (notably around the New Year period) imperfectly accounted for.

7. Observational Nature and Causal Identification

Most fundamentally, the study employs an observational time-series design rather than a randomized experimental framework. Although the controls and robustness checks substantially increase confidence in the observed patterns, unobserved confounding cannot be completely ruled out. The findings should therefore be interpreted as suggestive of the trial's early-phase impact rather than a definitive causal estimate.

Conclusion

Looking only at the raw numbers, curtailment appears to have gone up since the §13k trial began which, taken at face value, would suggest the policy isn't working. That impression turns out to be misleading. Once the analysis accounts for two other things happening at the same time Germany's continued build-out of wind and solar capacity, and the shutdown of its last nuclear power plants in April 2023 the apparent link between the trial and rising curtailment disappears entirely. In other words, curtailment rose for reasons that have nothing to do with §13k, and once those reasons are accounted for, there is no reliable evidence that the trial has changed curtailment levels in either direction so far.

This is not the same as concluding the policy has failed. As an additional check, the same analytical approach was tested on several dates where nothing policy-related actually happened, to see whether it would falsely detect an "effect" anyway and in a couple of cases, it did. That result is a useful reminder that roughly two years of trial data, covering a program that is still gradually signing up participants, is a thin basis for a firm verdict either way. A mechanism built around voluntary registration typically takes time to reach meaningful scale, and the trial itself is still running as of this report.

The honest summary is this: **Germany's curtailment problem kept growing for reasons unrelated to the §13k trial, and there is currently no solid evidence the trial has bent that trend one way or the other** but there also isn't enough data yet to rule out a real effect that simply hasn't shown up strongly enough to detect. The most useful next step is not a different statistical technique but more time: as participation in the program grows and more months of data accumulate, particularly once the trial concludes in September 2026 and transitions to a full competitive-tender system, a clearer picture should emerge. Until then, this report's contribution is less a verdict on §13k and more a demonstration of why verdicts on early-stage policy trials deserve real scrutiny before anyone inside or outside government treats a rising or falling number as proof of anything.

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The statistical analyses were run in Python, R and visualized with assistance from Google's Gemini AI